A Level • Cambridge (CIE) • Chemistry

43 mins 23 questions

Multiple Choice Practice

Topical Practice MCQ

18. Carboxylic acids and derivatives

Carefully selected past paper questions by topic.

37 P is a carboxylic acid with molecular formula $C_5H_{10}O_2$.

Carboxylic acid P reacts with an excess of $LiAlH_4$ to form compound Q.

Which pairs of molecules could be carboxylic acid P and compound Q?

	carboxylic acid P	compound Q
1	$CH_3(CH_2)_3COOH$	$CH_3(CH_2)_3OH$
2	$CH_3(CH_2)_3COOH$	$CH_3(CH_2)_3CHO$
3	$(CH_3)_3CCOOH$	$(CH_3)_3CCH_2OH$

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 only

35 Which substance reacts with ethanoic acid to give the organic product with the highest M_r?

A lithium aluminium hydride

B magnesium

C potassium carbonate

D propan-2-ol

37 A sample of propanoic acid of mass 3.70 g reacts with an excess of magnesium.

A second sample of propanoic acid of mass 3.70 g reacts with an excess of sodium.

Both reactions go to completion forming a gas.

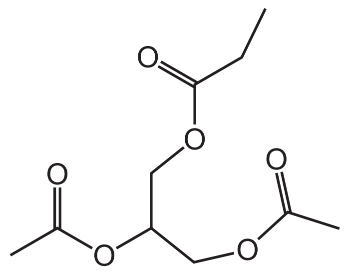
Which row is correct?

	volume of gas formed with magnesium at s.t.p. / cm^3	volume of gas formed with sodium at s.t.p. / cm^3
A	560	560
B	560	1120
C	1120	560
D	1120	1120

38 Two compounds, X and Y, are mixed and a little concentrated H_2SO_4 is added.

Ester Z is found in the resulting mixture of products.

ester Z



Which two compounds could be X and Y?

	X	Y
A	CH_3CH_2OH	$CH(CO_2H)_3$
B	CH_3CH_2OH	$CH_3CO_2CH_2CH(OH)CH_2OCOCH_2CH_3$
C	CH_3CO_2H	$CH_3CH_2CO_2CH_2CH(OH)CH_2OH$
D	CH_3CO_2H	$CH_2(OH)CH(OH)CH_2(OH)$

36 Compound Q can be hydrolysed by $HCl(aq)$. The two products of this hydrolysis have the same empirical formula.

What could Q be?

A $CH_3CO_2CH_2CH_2OH$

B $CH_3CO_2CH_2CH_2CO_2H$

C $CH_3CH_2CO_2CH_2CH_2CH_3$

D $CH_3CH_2CH(OH)CH(OH)CH_2CH_3$

38 How many esters with the molecular formula $C_5H_{10}O_2$ can be made by reacting a primary alcohol with a carboxylic acid?

A 4 **B** 5 **C** 6 **D** 8

38 Which compound can be used to make propanoic acid by treatment with a single reagent?

A $CH_2{=}CHCH_2CH_3$

B $CH_3CH_2CH_2CN$

C $CH_3CH(OH)CN$

D $CH_3CH(OH)CH_3$

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B $CH_3CO_2CH_2CH_2CO_2H$

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B $CH_3CH_2CH_2CN$

C $CH_3CH(OH)CN$

D $CH_3CH(OH)CH_3$

28 A carboxylic acid, P, has no possible chain isomers. It reacts with an alcohol, Q, that has only one positional isomer.

What could be the ester formed from a reaction between P and Q?

A butyl propanoate

B ethyl butanoate

C pentyl ethanoate

D propyl pentanoate

23 Carboxylic acids may be prepared by several different methods.

In which reaction would propanoic acid be formed?

A adding ammonium propanoate to dilute sulfuric acid

B heating ethyl propanoate with aqueous sodium hydroxide

C heating propan-2-ol with acidified potassium manganate(VII) under reflux

D heating propyl ethanoate with dilute sulfuric acid

26 Ethane-1,2-diol, $HOCH_2CH_2OH$, reacts with an excess of ethanoic acid, CH_3CO_2H, in the presence of an acid catalyst. A compound is formed with the molecular formula $C_6H_{10}O_4$.

What is the structure of this compound?

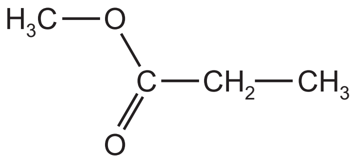
A $CH_3OCOCH_2CH_2CO_2CH_3$

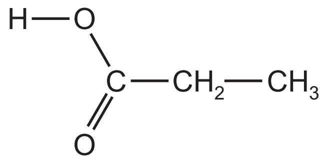
B $CH_3CO_2CH_2CH_2CO_2CH_3$

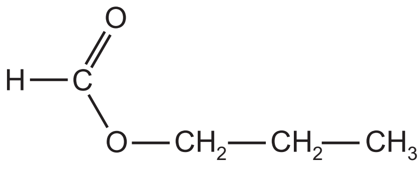
C $CH_3CO_2CH_2CH_2OCOCH_3$

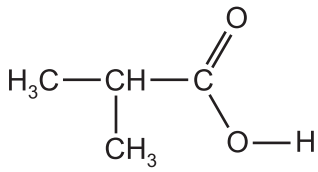
D $HOCH_2CH_2COCH_2OCOCH_3$

29 How many of the compounds shown will react with aqueous sodium hydroxide to form the sodium salt of a carboxylic acid?



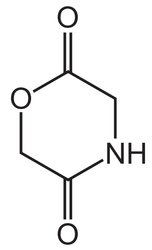






A 1 **B** 2 **C** 3 **D** 4

21 The cyclic compound M is heated with dilute hydrochloric acid.



compound M

What are the products of the reaction?

A $HOCH_2CO_2H$ and $H_2NCH_2CO_2H$

B HO_2CCH_2OH and $HO_2CCH_2NH_3^+$

C H_2NCOCH_2OH and $HOCH_2CHO$

D $HOCH_2CONH_3^+$ and $HOCH_2CHO$

20 Diesters can be made from diacids such as propane-1,3-dioic acid, $HO_2CCH_2CO_2H$.

Which combination of reactants would form the diester $CH_3CH_2OCOCH_2CH_2CO_2CH_2CH_3$?

A butane-1,4-dioic acid and ethanol

B ethanedioic acid and propan-1-ol

C ethanedioic acid, ethanol and butan-1-ol

D propane-1,3-dioic acid, ethanol and propan-1-ol

21 Lactic acid (2-hydroxypropanoic acid), $CH_3CH(OH)CO_2H$, is found in sour milk.

Which reaction could occur with lactic acid?

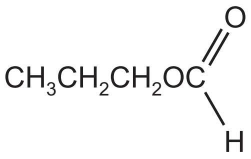
A $CH_3CH(OH)CO_2H + CH_3OH \rightarrow CH_3CH(OCH_3)CO_2H + H_2O$

B $CH_3CH(OH)CO_2H + HCO_2H \rightarrow CH_3CH(O_2CH)CO_2H + H_2O$

C $CH_3CH(OH)CO_2H + NaHCO_3 \rightarrow CH_3CH(ONa)CO_2H + H_2O + CO_2$

D $CH_3CH(OH)CO_2H + Cl_2 \rightarrow CH_3CH(Cl)CO_2H + HOCl$

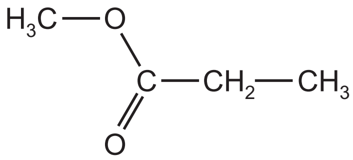
28 The structural formula of a compound **X** is shown below.

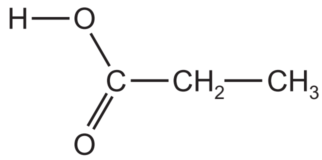


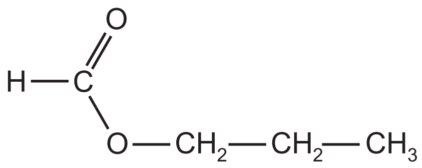
What is the name of compound **X** and how does its boiling point compare with that of butanoic acid?

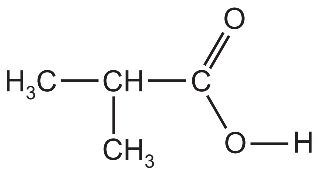
	name of **X**	boiling point of **X**
A	methyl propanoate	higher
B	methyl propanoate	lower
C	propyl methanoate	higher
D	propyl methanoate	lower

29 How many of the compounds shown will react with aqueous sodium hydroxide to form the sodium salt of a carboxylic acid?









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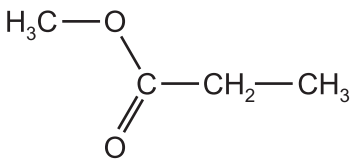
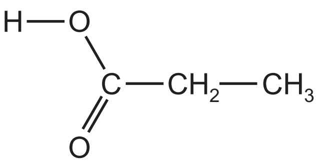
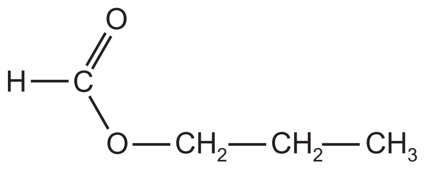
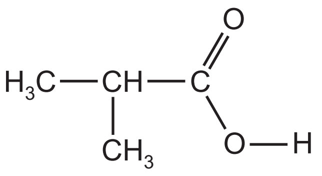


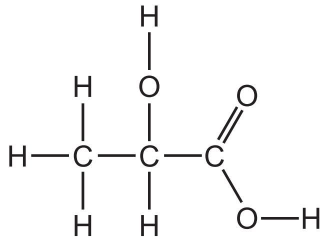






A 1 **B** 2 **C** 3 **D** 4

26 Lactic acid occurs naturally, for example in sour milk.



lactic acid

What is a property of lactic acid?

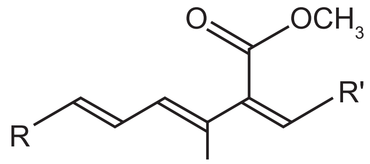
A It decolourises aqueous bromine rapidly.

B It is insoluble in water.

C It reduces Fehling's reagent.

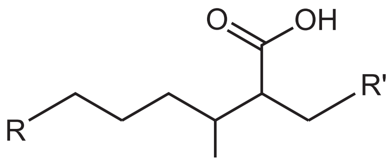
D Two molecules react with each other in the presence of a strong acid.

27 Part of the structure of strobilurin, a fungicide, is shown. R and R' are inert groups.



If strobilurin is first warmed with aqueous sulfuric acid, and its product then treated with hydrogen in the presence of a palladium catalyst, what could be the structure of the final product?

A



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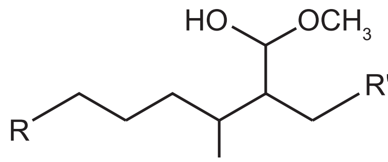
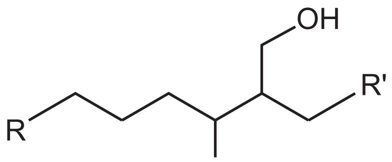
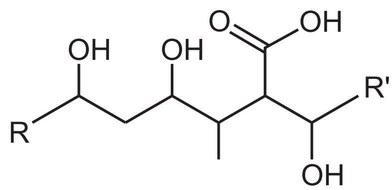


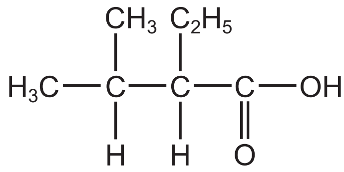
C



D

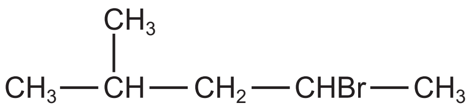


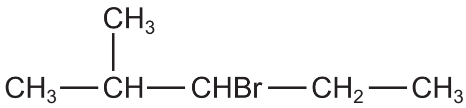
29 The characteristic odour of rum is attributed to the compound 2-ethyl-3-methylbutanoic acid.

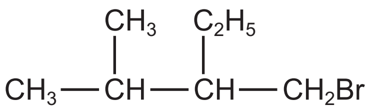


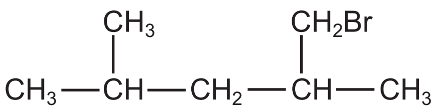
2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid by heating under reflux with alcoholic sodium cyanide and subsequent acid hydrolysis of the reaction product?

A 

B 

C 

D 

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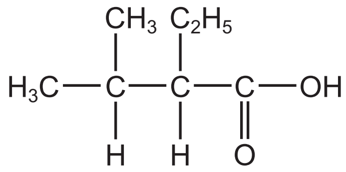

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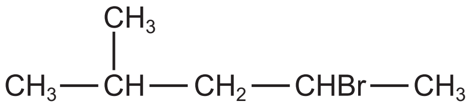
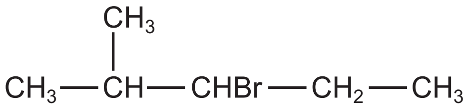
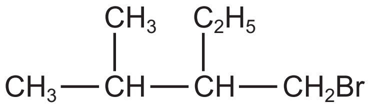
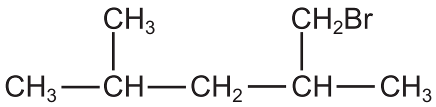
A 

B 

C 

D 

Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_11	18. Carboxylic acids and derivative...	**D**
2	9701_s24_qp_13	18. Carboxylic acids and derivative...	**B**
3	9701_w24_qp_12	18. Carboxylic acids and derivative...	**A**
4	9701_s23_qp_11	18. Carboxylic acids and derivative...	**C**
5	9701_s23_qp_13	18. Carboxylic acids and derivative...	**B**
6	9701_w23_qp_12	18. Carboxylic acids and derivative...	**C**
7	9701_s22_qp_13	18. Carboxylic acids and derivative...	**A**
8	9701_w19_qp_12	18. Carboxylic acids and derivative...	**B**
9	9701_w18_qp_12	18. Carboxylic acids and derivative...	**A**
10	9701_s17_qp_12	18. Carboxylic acids and derivative...	**A**
11	9701_s17_qp_13	18. Carboxylic acids and derivative...	**A**
12	9701_w16_qp_12	18. Carboxylic acids and derivative...	**C**
13	9701_w16_qp_12	18. Carboxylic acids and derivative...	**D**
14	9701_s15_qp_11	18. Carboxylic acids and derivative...	**B**
15	9701_s14_qp_13	18. Carboxylic acids and derivative...	**A**
16	9701_s13_qp_11	18. Carboxylic acids and derivative...	**B**
17	9701_s13_qp_11	18. Carboxylic acids and derivative...	**D**
18	9701_w13_qp_11	18. Carboxylic acids and derivative...	**D**
19	9701_w13_qp_12	18. Carboxylic acids and derivative...	**D**
20	9701_w10_qp_11	18. Carboxylic acids and derivative...	**D**
21	9701_w10_qp_12	18. Carboxylic acids and derivative...	**A**
22	9701_w09_qp_11	18. Carboxylic acids and derivative...	**B**
23	9701_w09_qp_12	18. Carboxylic acids and derivative...	**B**